

# Project Proposal

## GPU-Accelerated Batched Inference for Multilayer Perceptrons

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### Overview and Motivation

Neural network inference at scale is fundamentally a batched matrix computation problem. For a multilayer perceptron (MLP), each forward pass reduces to a sequence of matrix–vector multiplications and element-wise non-linearities. When many requests arrive simultaneously, these operations can be batched into matrix–matrix multiplications—a regime where GPU parallelism offers substantial throughput advantages over sequential CPU execution.

This project would implement batched MLP inference from scratch at the kernel level, without relying on high-level frameworks, and studies how different parallel execution strategies affect throughput and efficiency.

### Objectives

The project aims to:

- implement batched MLP forward pass in three progressively optimised configurations;
- benchmark each configuration across a range of batch sizes and model depths;
- analyse throughput, speedup, and memory bandwidth utilisation for each kernel;
- position results on a roofline model to characterise compute- vs. memory-bound regimes.

### Planned Implementations

1. **CPU baseline (C, sequential):** a straightforward implementation of batched matrix multiplication using row-major memory layout, serving as the reference for all speedup measurements.
2. **CUDA naïve kernel:** each output element is computed by a single thread with global memory accesses, exposing the cost of uncoalesced reads and the absence of data reuse.
3. **CUDA tiled kernel:** a shared-memory tiled matrix multiplication that loads tiles cooperatively into on-chip memory, reducing global memory traffic and improving arithmetic intensity.

All three implementations handle the same batched GEMM workload ( $\mathbf{Y} = \mathbf{XW}^\top + \mathbf{b}$ , repeated per layer) with ReLU activations, enabling a direct comparison.

### Evaluation

Experiments will sweep over batch sizes ( $B \in \{1, 8, 64, 256, 1024, 4096\}$ ) and network widths ( $d \in \{128, 512, 2048\}$ ) on a fixed machine. Measured quantities include:

- throughput (samples per second) and speedup relative to the CPU baseline;
- GPU kernel execution time (via CUDA events);

- effective memory bandwidth and arithmetic intensity;
- occupancy and shared memory utilisation.

## Technologies

- CUDA/C for GPU kernels (primary deliverable);
- C for the CPU baseline;
- CUDA profiling tools for performance analysis.

No external ML framework is used; all matrix operations are implemented directly. Synthetic random inputs are sufficient for all experiments.

## Expected Outcome

A well-documented CUDA/C implementation of batched MLP inference at three optimisation levels, accompanied by a written report presenting the experimental results, roofline analysis, and a discussion of the performance trade-offs between the naive and tiled GPU kernels.